

Safety Review Brief: Ozempic

Important Disclaimer

This report is not medical advice. FAERS disproportionality identifies reporting patterns, not incidence, patient risk, or causality.

Analysis Window

- Analysis period: 2025-04-01 to 2026-03-31
- All FAERS reports in period: 1,292,178
- Reports mentioning Ozempic: 20,245
- Drug-role scope: all mentions (not primary-suspect-only)
- PubMed drug terms: Ozempic, SEMAGLUTIDE, ORAL SEMAGLUTIDE, WEGOVY

Flagged Reporting Signals

Reaction	Cases	PRR (95% CI)	ROR (95% CI)	Retrieved / PubMed hits
cyclic vomiting syndrome	272	438.18 (313.35–612.73)	444.13 (317.42–621.42)	0 / 0
impaired gastric emptying	1,764	85.58 (79.77–91.82)	93.65 (87.05–100.76)	4 / 4
optic ischaemic neuropathy	389	73.84 (63.81–85.43)	75.26 (64.96–87.20)	1 / 1
gastrointestinal hypomotility	245	61.57 (51.65–73.39)	62.31 (52.22–74.36)	0 / 0
counterfeit product administered	128	45.43 (36.21–57.01)	45.72 (36.40–57.41)	0 / 0
weight loss poor	214	40.38 (34.02–47.91)	40.80 (34.34–48.47)	0 / 0
ileus	335	31.75 (27.86–36.18)	32.26 (28.27–36.82)	5 / 6
product label confusion	157	25.29 (21.03–30.42)	25.48 (21.16–30.68)	0 / 0
food craving	142	17.99 (14.93–21.66)	18.11 (15.02–21.83)	3 / 3
intestinal obstruction	499	13.30 (12.08–14.63)	13.61 (12.34–15.00)	5 / 8

Signal Details

cyclic vomiting syndrome

- Drug + event reports: 272
- Drug + other-event reports: 19,973
- Other-drug + event reports: 39
- Other-drug + other-event reports: 1,271,894
- PRR: 438.18 (95% CI 313.35–612.73)
- ROR: 444.13 (95% CI 317.42–621.42)
- Threshold evaluation:
- Case count 272 meets minimum 5.
- PRR 438.18 meets minimum 2.00.
- ROR lower 95% confidence bound 317.42 exceeds 1.00.

Retrieved PubMed Evidence

- Total PubMed matches: 0
- Abstracts displayed: 0
- Ranking method: pubmed_relevance

- Synthesis status: no_abstracts
- No PubMed abstracts were returned for this drug/event query.

impaired gastric emptying

- Drug + event reports: 1,764
- Drug + other-event reports: 18,481
- Other-drug + event reports: 1,295
- Other-drug + other-event reports: 1,270,638
- PRR: 85.58 (95% CI 79.77–91.82)
- ROR: 93.65 (95% CI 87.05–100.76)
- Threshold evaluation:
- Case count 1764 meets minimum 5.
- PRR 85.58 meets minimum 2.00.
- ROR lower 95% confidence bound 87.05 exceeds 1.00.

Retrieved PubMed Evidence

- Total PubMed matches: 4
- Abstracts displayed: 4
- Ranking method: semantic_embedding
- Synthesis status: completed

Grounded summary (supports): Multiple pharmacovigilance studies analyzing large FDA Adverse Event Reporting System (FAERS) databases provide consistent evidence associating the glucagon-like peptide-1 receptor agonist (GLP-1RA) semaglutide, a component of Ozempic, with reports of impaired gastric emptying (IGE). Studies report IGE as a frequent adverse event among GLP-1RAs, with semaglutide showing among the highest reporting odds ratios. The risk of IGE appears highest early after treatment initiation and may vary by age, sex, and weight. Oral semaglutide specifically shows signals for gastrointestinal disorders including impaired gastric emptying. These findings are observational, based on reported adverse events, and do not establish causality but underline the importance of monitoring for gastric motility effects during treatment. (PMID 38776037 (<https://pubmed.ncbi.nlm.nih.gov/38776037/>), PMID 39578156 (<https://pubmed.ncbi.nlm.nih.gov/39578156/>), PMID 42232176 (<https://pubmed.ncbi.nlm.nih.gov/42232176/>), PMID 42284324 (<https://pubmed.ncbi.nlm.nih.gov/42284324/>))

- supports: GLP-1 receptor agonists, including semaglutide (Ozempic), are notably associated with reports of impaired gastric emptying in pharmacovigilance data from the FAERS database. (PMID 39578156 (<https://pubmed.ncbi.nlm.nih.gov/39578156/>), PMID 42232176 (<https://pubmed.ncbi.nlm.nih.gov/42232176/>), PMID 42284324 (<https://pubmed.ncbi.nlm.nih.gov/42284324/>))
- supports: Risk factors such as younger age, female sex, and early treatment duration influence the reporting pattern of impaired gastric emptying events with semaglutide and other GLP-1RAs. (PMID 39578156 (<https://pubmed.ncbi.nlm.nih.gov/39578156/>), PMID 42232176 (<https://pubmed.ncbi.nlm.nih.gov/42232176/>))
- supports: Oral semaglutide shows consistent signals for gastrointestinal adverse events in FAERS data, including impaired gastric emptying, among other gastrointestinal signals. (PMID 38776037 (<https://pubmed.ncbi.nlm.nih.gov/38776037/>))

Evidence limitations:

- All evidence is derived from observational pharmacovigilance data, which may be subject to reporting biases and do not prove causality.
- Data are based on reported adverse events and lack detailed clinical confirmation or mechanistic insight.
- The analyses cover multiple GLP-1RAs but do not always separate effects of semaglutide specifically from other drugs in this class.
- Risk estimates are relative reporting odds and do not directly estimate incidence or absolute risk.
- Glucagon-like peptide-1 receptor agonists and impaired gastric emptying: a pharmacovigilance analysis of the US Food and Drug Administration adverse event reporting system. (British journal of anaesthesia; 2025 May; 10.1016/j.bja.2024.10.013) — PMID 39578156 (<https://pubmed.ncbi.nlm.nih.gov/39578156/>)
- Abstract: BACKGROUND: Glucagon-like peptide-1 receptor agonists (GLP-1RAs) potentially increase the risk of pulmonary aspiration resulting from impaired gastric emptying (IGE). We evaluated the association between GLP-1RAs and IGE using the US Food and Drug Administration Adverse Event Reporting System (FAERS).

METHODS: We analysed FAERS data from 2004 Q1 to 2024 Q1, identifying the top 10 drugs linked to IGE and determining the proportion of GLP-1RA use. Disproportionality analysis using the reporting odds ratio was conducted to assess the relative IGE risk for each drug. Logistic regression analysed the impact of age, weight, and sex on IGE risk. Cumulative incidence and time to onset of IGE events were examined using Kaplan-Meier and Weibull shape parameter tests. **RESULTS:** Among the top 10 drugs associated with IGE reports, five were GLP-1RAs, accounting for 49.5% (982/1982) of cases. Dulaglutide (odds ratio [OR] 0.97, 95% confidence interval [CI] 0.94-1.00, $P=0.033$) and semaglutide (OR 0.96, 95% CI 0.94-0.97, $P=0.001$) showed lower IGE risk with older age. For exenatide, higher weight (OR 0.99, 95% CI 0.98-1.00, $P=0.033$) and male sex (OR 0.39, 95% CI 0.20-0.68, $P=0.033$) were associated with lower IGE risk. Median onset times ranged from 40.5 days (semaglutide) to 107.5 days (tirzepatide) from initiation of therapy. The Weibull shape parameter β was <1 for all GLP-1RAs, indicating a higher IGE risk early in treatment. **CONCLUSIONS:** GLP-1RAs were notably associated with reports of impaired gastric emptying in the FAERS. Age, weight, and sex were significantly associated with impaired gastric emptying risk for certain GLP-1RAs. IGE events tended to occur early in treatment, with risk diminishing over time. These findings provide valuable references for future research on perioperative safety with GLP-1RAs.

- Risk factors for drug-related impaired gastric emptying: a pharmacovigilance analysis of the FDA adverse event reporting system. (Frontiers in pharmacology; 2026; 10.3389/fphar.2026.1844057) — PMID 42232176 (<https://pubmed.ncbi.nlm.nih.gov/42232176/>)
- Abstract: **BACKGROUND:** Pulmonary aspiration remains a major perioperative patient safety issue. Drug-related impaired gastric emptying (IGE) is a recognized yet underappreciated risk factor for aspiration. With the increasing use of medications such as glucagon-like peptide-1 receptor agonists (GLP-1RAs), concerns have grown regarding their potential to delay gastric emptying and thereby elevate aspiration risk. However, real-world data on drug-related IGE remain limited.

OBJECTIVE: To identify risk factors for drug-related IGE using real-world pharmacovigilance data from the FDA Adverse Event Reporting System (FAERS). **METHODS:** A retrospective pharmacovigilance study was conducted using FAERS data from Q1 2004 through Q2 2025. Disproportionality analysis, logistic regression, LASSO regression and time-to-onset analysis were performed to investigate the association between drugs and IGE from different perspectives. **RESULTS:** A total of 731 drugs related to IGE. Six of the top 10 most frequently reported drugs associated with IGE were antidiabetic agents, including five GLP-1RAs: semaglutide, dulaglutide, tirzepatide, exenatide, and liraglutide. Multi-factor analysis identified female sex, age <63 years, and exposure to any of nine specific drugs-including GLP-1RAs, insulin lispro, sodium oxybate, olmesartan, and esomeprazole-as independent risk factors for drug-related IGE. The model achieved an ROC-AUC of 0.739, indicating good discriminatory capability. Time-to-onset analysis revealed an early-failure pattern for all nine drugs. **CONCLUSION:** Our study shows that female sex, younger age, and use of specific medications are associated with an increased risk of drug-related IGE. These findings may provide valuable evidence to support the early recognition of drug-related IGE and for optimizing anaesthetic plans and perioperative personalized fasting strategies.

- Drug-induced gastric motility disorders: A disproportionality analysis from the FAERS and CVARD databases. (PloS one; 2026; 10.1371/journal.pone.0351731) — PMID 42284324 (<https://pubmed.ncbi.nlm.nih.gov/42284324/>)
- Abstract: **BACKGROUND:** Delayed gastric emptying and gastroesophageal reflux represent critical yet frequently underrecognized complications in hospitalized patients, particularly in the context of polypharmacy. While multiple medication classes have been implicated in disrupting gastrointestinal motility, the comprehensive risk spectrum of individual drugs remains poorly characterized. This study aimed to conduct a comprehensive disproportionality analysis to identify drugs associated with delayed gastric emptying and reflux using large-scale pharmacovigilance data.

METHODS: We analyzed adverse event reports from the FDA Adverse Event Reporting System (FAERS; 2004-2025; $n > 58$ million) and validated findings against the Canada Vigilance Adverse Reaction Online Database (CVARD). Disproportionality analysis was performed using Reporting Odds Ratio (ROR), Proportional Reporting Ratio (PRR), and Bayesian Confidence Propagation Neural Network (BCPNN). Weibull time-to-onset analysis was conducted to characterize temporal patterns of adverse event onset. **RESULTS:** Among the top 50 drugs screened, 20 demonstrated positive signals across all three algorithms. Glucagon-like peptide-1 (GLP-1) receptor agonists exhibited the strongest associations with gastric motility disorders, with semaglutide showing the highest ROR for impaired gastric emptying (ROR: 80.27; 95% CI: 76.39-84.34), validated in CVARD (ROR: 54.17). Insulin formulations, particularly insulin degludec (ROR: 18.90), bisphosphonates, angiotensin receptor blockers, and trofinetide also demonstrated significant signals. Weibull analysis revealed divergent temporal patterns, ranging from early-onset (trofinetide: median 6.6 days) to late-onset (immunoglobulin G: median 535.1 days). **CONCLUSION:** This study identifies a broad spectrum of drug-associated gastric motility disorders with distinct temporal profiles. These findings provide evidence-based priorities for enhanced pharmacovigilance and inform clinical decision-making to mitigate this preventable cause of morbidity.

- Adverse Events of Oral GLP-1 Receptor Agonist (Semaglutide Tablets): A Real-World Study Based on FAERS from 2019 to 2023. (Diabetes therapy : research, treatment and education of diabetes and related disorders; 2024 Aug; 10.1007/s13300-024-01594-7) — PMID 38776037 (<https://pubmed.ncbi.nlm.nih.gov/38776037/>)
- Abstract: **INTRODUCTION:** Glucagon-like peptide-1 receptor agonists (GLP-1RAs) have attracted much attention because of their significant hypoglycemic and weight-loss effects. Previous preparations can only be subcutaneously injected. Oral administration of GLP-1RAs semaglutide helps to broaden treatment options, but its safety in the real world still needs to be observed. This study is based on FDA adverse event reporting system (FAERS) database to mine adverse drug events (ADE) of oral semaglutide, and provide references for the clinical safe use of this drug.

METHODS: To analyze the signal quality of oral semaglutide, which is a drug used in the FAERS database from the third quarter of 2019 to the third quarter of 2023, we collected ADE data and performed data mining by using disproportionate analysis. Then, we standardized

the data and used a variety of signal-quantification techniques, including reported odds ratio (ROR), proportional reporting ratio (PRR), Bayesian belief propagation neural network (BCPNN), and multiple empirical Bayesian gamma Poisson contractions (MGPS), for further analysis. RESULTS: We screened 2398 reports on the use of semaglutide tablets, involving a total of 5653 ADE. These reports were mainly submitted by consumers, and the reporting country was mainly the United States. A total of 23 system organ classes (SOC) and 93 preferred terms (PT) were mined for the signals of semaglutide tablets. The three most common SOC were gastrointestinal disorders, general disorders and administration site conditions, and investigations. At the PT level, metabolism and nutrition disorders exhibit the highest number of signals, with the top three being thyroid cyst, acute cholecystitis, and ketosis. Gastrointestinal disorders rank second, primarily involving eructation, pancreatitis, impaired gastric emptying, and regurgitation. In addition, with nerve paralysis occurs and the signal intensity is high. CONCLUSIONS: Our study provides a deeper and broader understanding of the safety of oral semaglutide. The results of the ROR, PRR, BCPNN, and MGPS algorithms exhibit high consistency, with metabolism and nutrition-related disorders having the highest number of signals. The conclusions align with the technical specifications of the product. Notably, other unexpected effects are reported, including acute cholecystitis, paralysis of the abducens nerve, and positional vertigo.

optic ischaemic neuropathy

- Drug + event reports: 389
- Drug + other-event reports: 19,856
- Other-drug + event reports: 331
- Other-drug + other-event reports: 1,271,602
- PRR: 73.84 (95% CI 63.81–85.43)
- ROR: 75.26 (95% CI 64.96–87.20)
- Threshold evaluation:
- Case count 389 meets minimum 5.
- PRR 73.84 meets minimum 2.00.
- ROR lower 95% confidence bound 64.96 exceeds 1.00.

Retrieved PubMed Evidence

- Total PubMed matches: 1
- Abstracts displayed: 1
- Ranking method: semantic_embedding
- Synthesis status: completed

Grounded summary (supports): A pharmacovigilance study using the FDA Adverse Event Reporting System found significant safety signals for optic ischaemic neuropathy associated with semaglutide (Ozempic), as well as tirzepatide and liraglutide. Semaglutide showed a strong association with this adverse event based on disproportionality analysis. No such signal was found for other related drugs dulaglutide, exenatide, or lixisenatide. These findings are derived from post-marketing reports and suggest a potential safety concern with semaglutide, though causality has not been established and further prospective studies with confirmed diagnoses are recommended. (PMID 42259643 (<https://pubmed.ncbi.nlm.nih.gov/42259643/>))

- supports: Semaglutide (Ozempic) demonstrates a significant post-marketing safety signal for optic ischaemic neuropathy based on disproportionality analysis of adverse event reports. (PMID 42259643 (<https://pubmed.ncbi.nlm.nih.gov/42259643/>))
- supports: Tirzepatide and liraglutide also show significant safety signals for optic ischaemic neuropathy. (PMID 42259643 (<https://pubmed.ncbi.nlm.nih.gov/42259643/>))
- supports: No safety signal for optic ischaemic neuropathy was identified with dulaglutide, exenatide, or lixisenatide. (PMID 42259643 (<https://pubmed.ncbi.nlm.nih.gov/42259643/>))

Evidence limitations:

- Data are derived from spontaneous adverse event reports, which may be subject to reporting bias and cannot establish causality.
- The diagnoses of optic ischaemic neuropathy in these reports were not confirmed by neuro-ophthalmologists.
- Absolute risk estimates cannot be determined from disproportionality analyses.
- Findings pertain to post-marketing surveillance and require confirmation in prospective studies.
- Optic Ischaemic Neuropathy in Incretin-Based Therapy: A Comparative Analysis of Real-World Safety Data. (Diabetes, obesity & metabolism; 2026 Jun 08; 10.1111/dom.70952) — PMID 42259643 (<https://pubmed.ncbi.nlm.nih.gov/42259643/>)

- Abstract: AIMS: Non-arteritic anterior ischemic optic neuropathy (NAION) has emerged as a safety concern with semaglutide, prompting formal regulatory review and action by the European Medicines Agency. However, its occurrence across the broader class of incretin-based therapies is insufficiently characterised. This study used the FDA Adverse Event Reporting System to evaluate and compare pharmacovigilance signals for optic ischaemic neuropathy across six incretin-based therapies.

MATERIALS AND METHODS: Adverse event reports from each drug's approval date through Q3 2025 were extracted using OpenVigil 2.1. The Medical Dictionary for Regulatory Activities term "optic ischaemic neuropathy" was used as the outcome. Disproportionality analysis was performed using reporting odds ratio (ROR) and proportional reporting ratio (PRR) with 95% confidence intervals, applying Evans' criteria, restricted to primary suspect drug reports. **RESULTS:** Semaglutide demonstrated a strong post-marketing safety signal for optic ischaemic neuropathy (355 reports; ROR 94.45, 95% CI: 83.02-107.45; PRR 93.77, 95% CI: 82.48-106.61). Tirzepatide and liraglutide also exhibited significant safety signals (49 reports; ROR 2.94, 95% CI: 2.20-3.93; PRR 2.94, 95% CI: 2.20-3.93; and 13 reports; ROR 4.58, 95% CI: 2.65-7.91; PRR 4.58, 95% CI: 2.65-7.91, respectively). No signal was identified for dulaglutide, exenatide, or lixisenatide. Among semaglutide-associated reports, mean age was 59 years; disability was documented in 18% and hospitalization in 11% of cases. **CONCLUSIONS:** Significant post-marketing safety signals for optic ischaemic neuropathy were identified for semaglutide, tirzepatide, and liraglutide. The absence of signals with other glucagon-like peptide-1 receptor agonists argues against a uniform class effect. Prospective studies with neuro-ophthalmologist-confirmed diagnoses are warranted to establish causality and quantify absolute risk.

gastrointestinal hypomotility

- Drug + event reports: 245
- Drug + other-event reports: 20,000
- Other-drug + event reports: 250
- Other-drug + other-event reports: 1,271,683
- PRR: 61.57 (95% CI 51.65–73.39)
- ROR: 62.31 (95% CI 52.22–74.36)
- Threshold evaluation:
- Case count 245 meets minimum 5.
- PRR 61.57 meets minimum 2.00.
- ROR lower 95% confidence bound 52.22 exceeds 1.00.

Retrieved PubMed Evidence

- Total PubMed matches: 0
- Abstracts displayed: 0
- Ranking method: pubmed_relevance
- Synthesis status: no_abstracts
- No PubMed abstracts were returned for this drug/event query.

counterfeit product administered

- Drug + event reports: 128
- Drug + other-event reports: 20,117
- Other-drug + event reports: 177
- Other-drug + other-event reports: 1,271,756
- PRR: 45.43 (95% CI 36.21–57.01)
- ROR: 45.72 (95% CI 36.40–57.41)
- Threshold evaluation:
- Case count 128 meets minimum 5.
- PRR 45.43 meets minimum 2.00.
- ROR lower 95% confidence bound 36.40 exceeds 1.00.

Retrieved PubMed Evidence

- Total PubMed matches: 0
- Abstracts displayed: 0
- Ranking method: pubmed_relevance
- Synthesis status: no_abstracts
- No PubMed abstracts were returned for this drug/event query.

weight loss poor

- Drug + event reports: 214
- Drug + other-event reports: 20,031
- Other-drug + event reports: 333
- Other-drug + other-event reports: 1,271,600
- PRR: 40.38 (95% CI 34.02–47.91)
- ROR: 40.80 (95% CI 34.34–48.47)
- Threshold evaluation:
- Case count 214 meets minimum 5.
- PRR 40.38 meets minimum 2.00.
- ROR lower 95% confidence bound 34.34 exceeds 1.00.

Retrieved PubMed Evidence

- Total PubMed matches: 0
- Abstracts displayed: 0
- Ranking method: pubmed_relevance
- Synthesis status: no_abstracts
- No PubMed abstracts were returned for this drug/event query.

ileus

- Drug + event reports: 335
- Drug + other-event reports: 19,910
- Other-drug + event reports: 663
- Other-drug + other-event reports: 1,271,270
- PRR: 31.75 (95% CI 27.86–36.18)
- ROR: 32.26 (95% CI 28.27–36.82)
- Threshold evaluation:
- Case count 335 meets minimum 5.
- PRR 31.75 meets minimum 2.00.
- ROR lower 95% confidence bound 28.27 exceeds 1.00.

Retrieved PubMed Evidence

- Total PubMed matches: 6
- Abstracts displayed: 5
- Ranking method: semantic_embedding
- Synthesis status: completed

Grounded summary (mixed): The evidence regarding the association between Ozempic (semaglutide) and ileus or bowel obstruction is mixed. A systematic review and meta-analysis of GLP-1 receptor agonists (which includes semaglutide) found no significant overall increased risk of bowel obstruction or ileus; however, a higher risk was noted specifically with liraglutide, not semaglutide. Several case

reports document instances of ileus or small bowel obstruction temporally associated with semaglutide use, including serious complications such as acute kidney injury, suggesting a possible, though rare, adverse effect. Conversely, a retrospective cohort study in patients with inflammatory bowel disease found no increase in gastrointestinal adverse events, including ileus, with GLP-1 RA therapy (majority including semaglutide). Thus, while population-level data do not show a significant risk increase with semaglutide, isolated case reports indicate potential rare occurrences of ileus and bowel obstruction. (PMID 39469548 (<https://pubmed.ncbi.nlm.nih.gov/39469548/>), PMID 39964295 (<https://pubmed.ncbi.nlm.nih.gov/39964295/>), PMID 40830314 (<https://pubmed.ncbi.nlm.nih.gov/40830314/>), PMID 41294925 (<https://pubmed.ncbi.nlm.nih.gov/41294925/>), PMID 41913883 (<https://pubmed.ncbi.nlm.nih.gov/41913883/>))

- supports: GLP-1 receptor agonists as a class do not significantly increase the risk of ileus or bowel obstruction overall, but liraglutide shows a higher risk than semaglutide. (PMID 39964295 (<https://pubmed.ncbi.nlm.nih.gov/39964295/>))
- supports: Case reports describe ileus or small bowel obstruction occurring in patients using semaglutide, sometimes with serious complications such as acute kidney injury. (PMID 41294925 (<https://pubmed.ncbi.nlm.nih.gov/41294925/>), PMID 39469548 (<https://pubmed.ncbi.nlm.nih.gov/39469548/>), PMID 41913883 (<https://pubmed.ncbi.nlm.nih.gov/41913883/>))
- supports: In patients with inflammatory bowel disease, GLP-1 receptor agonist therapy including semaglutide does not increase gastrointestinal adverse events such as ileus over 12 months. (PMID 40830314 (<https://pubmed.ncbi.nlm.nih.gov/40830314/>))

Evidence limitations:

- Meta-analysis showed high heterogeneity and included various GLP-1 RAs, limiting specificity to semaglutide.
- Case reports are observational and cannot establish causality, reflecting rare events.
- No randomized controlled trials focusing exclusively on semaglutide and ileus risk were identified.
- Data on off-label high-dose or unsupervised semaglutide use is very limited.
- Evaluating bowel obstruction and ileus events in patients on GLP-1 receptor agonists: a systematic review and meta-analysis. (Expert opinion on drug safety; 2026 Jun; 10.1080/14740338.2025.2465870) — PMID 39964295 (<https://pubmed.ncbi.nlm.nih.gov/39964295/>)
- Abstract: INTRODUCTION: This research evaluated the association between glucagon-like peptide-1 receptor agonists (GLP-1 RA) and bowel obstruction or ileus.

METHODS: We searched databases, including Medline, Embase, and Cochrane, for studies on adult patients treated with GLP-1 RA. We included randomized control trials (RCT), cohort, case-control studies, and case reports. We used the Cochrane Risk of Bias tool to evaluate the quality of RCTs and the Newcastle-Ottawa Scale for cohort and case-control studies. **RESULTS:** Out of 317 records identified, 14 studies were included in the systematic review. The meta-analysis included 6 studies with a combined total of 550,426 participants. The use of GLP-1 RA did not show an incremental risk of bowel obstruction or ileus compared to controls (OR 1.95, 95% CI 0.43-8.79). However, the studies had high heterogeneity ($I^2 = 94\%$). A subgroup analysis by specific medication revealed that liraglutide was associated with a significantly high risk of bowel obstruction or ileus (OR 3.0, 95% CI 2.03-4.45; $I^2 = 0\%$). **CONCLUSIONS:** GLP-1 receptor agonists do not significantly increase the risk of bowel obstruction or ileus. However, liraglutide is associated with a higher risk compared to semaglutide. Clinicians should remain aware of these rare events while recognizing the benefits of GLP-1 receptor agonists for glycemic control and cardiovascular health. **PROSPERO REGISTRATION:** CRD42024585971.

- Subocclusive ileus following off-label high-dose semaglutide use. (Endocrinology, diabetes & metabolism case reports; 2025 Oct 01; 10.1530/EDM-25-0081) — PMID 41294925 (<https://pubmed.ncbi.nlm.nih.gov/41294925/>)
- Abstract: SUMMARY: Semaglutide, a GLP-1 receptor agonist (RA), is approved for the treatment of type 2 diabetes and obesity. However, its increasing off-label use, particularly for cosmetic weight loss, raises concerns considering its potential adverse effects. We report the case of a 37-year-old woman with normal body mass index who developed subocclusive ileus following unsupervised use of semaglutide.

LEARNING POINTS: Our case highlights that semaglutide can lead to functional intestinal obstruction. Clinicians should maintain a high index of suspicion for GLP-1 RA related gastrointestinal complications.

- Glucagon-Like Peptide-1 Receptor Agonist Therapy Does Not Increase Gastrointestinal Adverse Events in Patients with Inflammatory Bowel Disease. (Digestive diseases and sciences; 2025 Oct; 10.1007/s10620-025-09344-w) — PMID 40830314 (<https://pubmed.ncbi.nlm.nih.gov/40830314/>)
- Abstract: BACKGROUND: As the prevalence of obesity rises among patients with inflammatory bowel disease (IBD), concerns have emerged regarding the gastrointestinal safety of glucagon-like peptide-1 receptor agonists (GLP-1 RAs), which are increasingly used for diabetes and weight management.

METHODS: We conducted a retrospective cohort study of adult patients with IBD on GLP-1 RA therapy within a large academic medical system between 2010 and 2024. We compared gastrointestinal adverse events-including ileus or bowel obstruction, bowel surgery, IBD-related hospitalization, and escalation of medical therapy-occurring in the one-year period preceding and following GLP-1 RA exposure, which was defined by at least two prescriptions within 90 days. **RESULTS:** Among 271 patients (median age 62.8 years; 62% female; 64% ulcerative colitis), 80 completed 12 months and an additional 168 completed 6 months of GLP-1 RA therapy. Semaglutide was the most common agent (58%), followed by dulaglutide (28%), tirzepatide (14%), and liraglutide (1%). No significant difference in gastrointestinal adverse event rates was observed between pre- and post-treatment intervals when normalized for exposure time ($p \geq 0.10$)

for all comparable adverse events). Median BMI declined modestly from 33.8 to 33.4 kg/m² at 12 months ($p = 0.20$). Subgroup analyses by agent, sex, IBD type, and GLP-1 RA indication yielded similar findings. **CONCLUSIONS:** GLP-1 RA therapy does not increase gastrointestinal adverse events in patients with IBD over 12 months, supporting its safety in this population.

- Food-induced small bowel obstruction observed in a patient with inappropriate use of semaglutide. (Diabetology international; 2024 Oct; 10.1007/s13340-024-00751-4) — PMID 39469548 (<https://pubmed.ncbi.nlm.nih.gov/39469548/>)
- Abstract: We herein report a case of food-induced small bowel obstruction (FIBO) while using a glucagon-like peptide 1 receptor agonist (GLP1-RA), trying to lose weight due to distorted body image. The patient was a 30-year-old woman who was not obese (height 158 cm, weight 50 kg). She started taking an oral semaglutide, a GLP1-RA, and it was soon switched to weekly subcutaneous semaglutide because of ineffectiveness. More than 6 months after titrating up to 1.0 mg, she got drunk and chomped on a lot of scallops sandwiched between sheets of kelp, so-called "kobujime" in Japan, and half a day later complained of abdominal pain. Based on a finding of computed tomography at our emergency department, she was suspected of having a bowel obstruction and underwent laparoscopic surgery, which resulted in a diagnosis of small bowel obstruction by kelp. FIBO is rare, but it can become very severe once it happens. Although we cannot prove the direct pathophysiological effects of GLP1-RAs on FIBO in this particular case, GLP1-RAs have been reported to be one of the underlying risks of bowel obstruction based on epidemiological and basic research evidence; still, it is under-recognized. For example, the package inserts in Japan do not mention intestinal obstruction. We hope that the present report will prove helpful in paying attention to GLP1-RAs as a factor in bowel obstruction, including FIBO.
- A Rare Case of Semaglutide-Associated Small Bowel Obstruction Complicated by Acute Kidney Injury Requiring Dialysis. (Cureus; 2026 Feb; 10.7759/cureus.104376) — PMID 41913883 (<https://pubmed.ncbi.nlm.nih.gov/41913883/>)
- Abstract: Glucagon-like peptide-1 (GLP-1) receptor agonists are widely used in the treatment of diabetes mellitus and obesity. They have proven to be cardioprotective, neuroprotective, and renoprotective, and are now also being considered for drug addiction. There has been increasing demand for these medications over the last several years, with the most common side effects being nausea, vomiting, and constipation. Nausea and vomiting are known to improve over a period of time due to tachyphylaxis. We present the case of a 59-year-old Hispanic obese female on semaglutide for weight loss. Her medication dose was recently increased from 0.25 mg to 0.5mg under her physician's supervision. She presented with nausea, vomiting, and abdominal pain, which was later diagnosed as small bowel obstruction (SBO) from severe ileus, and developed acute kidney injury (AKI) requiring hemodialysis for several weeks. This case was also complicated because of the past medical history of autoimmune disorder and granulomatosis polyangiitis, which required additional testing to rule out a possible autoimmune etiology of the current presentation. To our knowledge, this is a rare case of semaglutide-associated SBO and concomitant AKI requiring temporary dialysis, highlighting the adverse effects of GLP-1 receptor agonists and the need for further studies to help formulate best practice guidelines.

product label confusion

- Drug + event reports: 157
- Drug + other-event reports: 20,088
- Other-drug + event reports: 390
- Other-drug + other-event reports: 1,271,543
- PRR: 25.29 (95% CI 21.03–30.42)
- ROR: 25.48 (95% CI 21.16–30.68)
- Threshold evaluation:
- Case count 157 meets minimum 5.
- PRR 25.29 meets minimum 2.00.
- ROR lower 95% confidence bound 21.16 exceeds 1.00.

Retrieved PubMed Evidence

- Total PubMed matches: 0
- Abstracts displayed: 0
- Ranking method: pubmed_relevance
- Synthesis status: no_abstracts
- No PubMed abstracts were returned for this drug/event query.

food craving

- Drug + event reports: 142
- Drug + other-event reports: 20,103

- Other-drug + event reports: 496
- Other-drug + other-event reports: 1,271,437
- PRR: 17.99 (95% CI 14.93–21.66)
- ROR: 18.11 (95% CI 15.02–21.83)
- Threshold evaluation:
- Case count 142 meets minimum 5.
- PRR 17.99 meets minimum 2.00.
- ROR lower 95% confidence bound 15.02 exceeds 1.00.

Retrieved PubMed Evidence

- Total PubMed matches: 3
- Abstracts displayed: 3
- Ranking method: semantic_embedding
- Synthesis status: completed

Grounded summary (supports): Semaglutide, including formulations such as once-weekly subcutaneous semaglutide (2.4 mg) and oral semaglutide, has been associated with reduced food cravings and changes in food preferences. A randomized controlled trial reported that weekly subcutaneous semaglutide decreased food cravings, improved control of eating, and reduced energy intake and appetite in adults with obesity (PMID 33269530). A retrospective multicenter study in Japanese patients with type 2 diabetes receiving oral semaglutide found decreased total energy intake and declines in cravings for sweet, chocolate-flavored, and starchy foods, alongside improved satisfaction at meals and reduced difficulty in resisting food cravings (PMID 39878627). A broader narrative review underscored the importance of addressing food cravings in medical weight management and noted that certain anti-obesity medications, including semaglutide, can reduce cravings (PMID 39408206). (PMID 33269530 (<https://pubmed.ncbi.nlm.nih.gov/33269530/>), PMID 39408206 (<https://pubmed.ncbi.nlm.nih.gov/39408206/>), PMID 39878627 (<https://pubmed.ncbi.nlm.nih.gov/39878627/>))

- supports: Once-weekly subcutaneous semaglutide 2.4 mg reduced food cravings and improved control of eating in adults with obesity. (PMID 33269530 (<https://pubmed.ncbi.nlm.nih.gov/33269530/>))
- supports: Oral semaglutide treatment decreased total energy intake and cravings for sweet, chocolate, and starchy foods in Japanese patients with type 2 diabetes. (PMID 39878627 (<https://pubmed.ncbi.nlm.nih.gov/39878627/>))
- supports: Certain anti-obesity medications, including semaglutide, have data indicating craving control efficacy, important for long-term weight loss maintenance. (PMID 39408206 (<https://pubmed.ncbi.nlm.nih.gov/39408206/>))

Evidence limitations:

- The evidence includes a randomized controlled trial and observational retrospective study, which may have differing levels of evidence strength.
- The retrospective study is limited to Japanese patients with type 2 diabetes and may have limited generalizability.
- The narrative review does not provide direct clinical trial data specifically for Ozempic but discusses the class and related medications broadly.
- Direct assessment of food craving as an adverse event was not the primary endpoint in these studies; findings are related to pharmacologic effects on appetite and cravings.
- Data are limited to semaglutide formulations and do not detail adverse event frequency or severity regarding food craving as a negative event.
- Tackling Cravings in Medical Weight Management: An Update on Pathophysiology and an Integrated Approach to Treatment. (Nutrients; 2024 Sep 25; 10.3390/nu16193238) — PMID 39408206 (<https://pubmed.ncbi.nlm.nih.gov/39408206/>)
- Abstract: Background/Objectives: Food cravings involve a strong drive to consume palatable foods irrespective of nutritional status. Importantly, cravings contribute substantially to the obesity epidemic. Managing hunger alone is insufficient for weight management as this relates only to homeostatic eating and does not address the complex aetiology of hedonic eating and its crucial role in food cravings. Medical weight management clinics and anti-obesity medication trials do not routinely identify and address food cravings. Methods: We conducted a narrative review of the literature consisting of 115 peer-reviewed articles (original articles and reviews). We included articles focused on food craving pathophysiology, assessment, and management strategies providing contrasts against the current medical model of weight management seen in obesity pharmacotherapy trials as well as the current standard of practise. Results: We outline the neurohormonal and psychological drivers of cravings, which lead to a spectrum of eating behaviours, from comfort food eating to binge eating disorders. We provide an overview of ways of identification and measurement options, including their strengths and weaknesses, and an overview of management strategies and their cravings control efficacy, spanning lifestyle

modifications like nutrition and sleep, psychological therapies (i.e., cognitive behavioural therapy [CBT], acceptance-based therapies such as mindfulness) and, last but not least, medications that not only are approved for weight reduction but reduce cravings. Finally, based on these findings, we provide a proposed integrated and iterative model that is able to evolve and adapt to the individual over time in tackling cravings for long-term weight loss maintenance. Conclusions: The findings emphasise the importance of cravings management and provide a synthesis on how cravings can be identified in a medical weight management setting, which can be practically implemented in an integrated iterative model spanning anti-obesity medications that have craving control data to evidence-based lifestyle and psychological interventions.

- The effect of semaglutide 2.4 mg once weekly on energy intake, appetite, control of eating, and gastric emptying in adults with obesity. (Diabetes, obesity & metabolism; 2021 Mar; 10.1111/dom.14280) — PMID 33269530 (<https://pubmed.ncbi.nlm.nih.gov/33269530/>)
- Abstract: AIM: To investigate the effects of once-weekly subcutaneous (s.c.) semaglutide 2.4 mg on gastric emptying, appetite, and energy intake in adults with obesity.

MATERIALS AND METHODS: A double-blind, parallel-group trial was conducted in 72 adults with obesity, randomized to once-weekly s.c. semaglutide (dose-escalated to 2.4 mg) or placebo for 20 weeks. Gastric emptying was assessed using paracetamol absorption following a standardized breakfast. Participant-reported appetite ratings and Control of Eating Questionnaire (CoEQ) responses were assessed, and energy intake was measured during ad libitum lunch. **RESULTS:** The area under the concentration-time curve (AUC) for paracetamol 0 to 5 hours after a standardized meal (AUC_{0-5h,para}; primary endpoint) was increased by 8% (P = 0.005) with semaglutide 2.4 mg versus placebo at week 20 (non-significant when corrected for week 20 body weight; P = 0.12). No effect was seen on AUC_{0-1h,para}, maximum observed paracetamol concentration, or time to maximum observed paracetamol concentration. Ad libitum energy intake was 35% lower with semaglutide versus placebo (1736 versus 2676 kJ; estimated treatment difference -940 kJ; P < 0.0001). Semaglutide reduced hunger and prospective food consumption, and increased fullness and satiety when compared with placebo (all P < 0.02). The CoEQ indicated better control of eating and fewer/weaker food cravings with semaglutide versus placebo (P < 0.05). Body weight was reduced by 9.9% with semaglutide and 0.4% with placebo. Safety was consistent with the known profile of semaglutide. **CONCLUSIONS:** In adults with obesity, once-weekly s.c. semaglutide 2.4 mg suppressed appetite, improved control of eating, and reduced food cravings, ad libitum energy intake and body weight versus placebo. There was no evidence of delayed gastric emptying at week 20, assessed indirectly via paracetamol absorption.

- Changes in food preferences after oral semaglutide administration in Japanese patients with type 2 diabetes: KAMOGAWA-DM cohort. (Diabetes & vascular disease research; 2025; 10.1177/14791641251318309) — PMID 39878627 (<https://pubmed.ncbi.nlm.nih.gov/39878627/>)
- Abstract: BACKGROUND: This study aimed to investigate the effects of oral semaglutide on the changes in food preference of Japanese patients with type 2 diabetes.

METHODS: This retrospective multicenter study included 75 patients with type 2 diabetes who received oral semaglutide. The primary outcome was the change in the score of brief-type self-administered diet history questionnaire (BDHQ) score 3 months after the initiation of oral semaglutide treatment. The secondary outcome was the change in the Control of Eating Questionnaire (CoEQ), HbA1c, and body mass index (BMI) after 3 months. **RESULTS:** The median age, BMI, and HbA1c of the 23 participants were 64.0 years, 26.9 kg/m², and 7.6% (59 mmol/mol). The BDHQ results showed total energy was significantly reduced. Among the individual nutrients, carbohydrates most decreased. The CoEQ results particularly showed declines in cravings for something sweet, chocolate or chocolate flavored foods, and starchy foods, satisfaction at meals, frequency and intensity of food craving, difficulty of resisting the craving for food, and frequency of eating in response to cravings for food were significantly lower after 3 months. The mean HbA1c and BMI significantly decreased. **CONCLUSIONS:** In Japanese patients with type 2 diabetes, oral semaglutide treatment decreased total energy intake and changed food preferences.

intestinal obstruction

- Drug + event reports: 499
- Drug + other-event reports: 19,746
- Other-drug + event reports: 2,358
- Other-drug + other-event reports: 1,269,575
- PRR: 13.30 (95% CI 12.08–14.63)
- ROR: 13.61 (95% CI 12.34–15.00)
- Threshold evaluation:
- Case count 499 meets minimum 5.
- PRR 13.30 meets minimum 2.00.
- ROR lower 95% confidence bound 12.34 exceeds 1.00.

Retrieved PubMed Evidence

- Total PubMed matches: 8
- Abstracts displayed: 5
- Ranking method: semantic_embedding
- Synthesis status: completed

Grounded summary (mixed): Evidence from multiple study types including a meta-analysis, case reports, and pharmacovigilance data have been reviewed regarding the association of semaglutide (a GLP-1 receptor agonist marketed as Ozempic) with intestinal obstruction. A meta-analysis pooling data on GLP-1 receptor agonists did not find a statistically significant overall increase in bowel obstruction risk, although liraglutide showed a significant association whereas semaglutide did not. Individual case reports describe instances of small bowel obstruction or subocclusive ileus in patients using semaglutide, often in contexts of off-label or inappropriate use, suggesting possible functional obstruction events. Pharmacovigilance analyses from FAERS and EudraVigilance databases reported signals for intestinal obstruction with semaglutide, indicating disproportionate reporting but not proving causality. Overall, the evidence is mixed and largely observational with some signal detection; no definitive increased risk of intestinal obstruction with semaglutide has been conclusively established but caution is advised especially in certain clinical contexts. (PMID 38943656

(<https://pubmed.ncbi.nlm.nih.gov/38943656/>), PMID 39469548 (<https://pubmed.ncbi.nlm.nih.gov/39469548/>), PMID 39964295 (<https://pubmed.ncbi.nlm.nih.gov/39964295/>), PMID 41294925 (<https://pubmed.ncbi.nlm.nih.gov/41294925/>), PMID 42356493 (<https://pubmed.ncbi.nlm.nih.gov/42356493/>))

- supports: GLP-1 receptor agonists overall do not significantly increase the risk of bowel obstruction or ileus, but liraglutide is associated with a higher risk compared to semaglutide. (PMID 39964295 (<https://pubmed.ncbi.nlm.nih.gov/39964295/>))
- supports: Case reports describe food-induced small bowel obstruction and subocclusive ileus in patients using semaglutide, indicating possible intestinal obstruction events in individual cases. (PMID 39469548 (<https://pubmed.ncbi.nlm.nih.gov/39469548/>), PMID 41294925 (<https://pubmed.ncbi.nlm.nih.gov/41294925/>))
- supports: Pharmacovigilance data analyses identified unexpected adverse event signals including intestinal obstruction associated with semaglutide use in post-marketing surveillance databases. (PMID 38943656 (<https://pubmed.ncbi.nlm.nih.gov/38943656/>), PMID 42356493 (<https://pubmed.ncbi.nlm.nih.gov/42356493/>))

Evidence limitations:

- Most evidence is observational, including case reports and pharmacovigilance data, which cannot establish causality.
- The meta-analysis included heterogeneous studies and showed high variability; semaglutide-specific data indicated no significant risk increase but sample sizes may be insufficient to detect rare events.
- Case reports often involve off-label or inappropriate semaglutide use, limiting generalizability.
- Pharmacovigilance signals reflect reporting disproportionality and may be influenced by reporting biases and lack of denominator data.
- Evaluating bowel obstruction and ileus events in patients on GLP-1 receptor agonists: a systematic review and meta-analysis. (Expert opinion on drug safety; 2026 Jun; 10.1080/14740338.2025.2465870) — PMID 39964295 (<https://pubmed.ncbi.nlm.nih.gov/39964295/>)
- Abstract: INTRODUCTION: This research evaluated the association between glucagon-like peptide-1 receptor agonists (GLP-1 RA) and bowel obstruction or ileus.

METHODS: We searched databases, including Medline, Embase, and Cochrane, for studies on adult patients treated with GLP-1 RA. We included randomized control trials (RCT), cohort, case-control studies, and case reports. We used the Cochrane Risk of Bias tool to evaluate the quality of RCTs and the Newcastle-Ottawa Scale for cohort and case-control studies. RESULTS: Out of 317 records identified, 14 studies were included in the systematic review. The meta-analysis included 6 studies with a combined total of 550,426 participants. The use of GLP-1 RA did not show an incremental risk of bowel obstruction or ileus compared to controls (OR 1.95, 95% CI 0.43-8.79). However, the studies had high heterogeneity ($I^2 = 94\%$). A subgroup analysis by specific medication revealed that liraglutide was associated with a significantly high risk of bowel obstruction or ileus (OR 3.0, 95% CI 2.03-4.45; $I^2 = 0\%$). CONCLUSIONS: GLP-1 receptor agonists do not significantly increase the risk of bowel obstruction or ileus. However, liraglutide is associated with a higher risk compared to semaglutide. Clinicians should remain aware of these rare events while recognizing the benefits of GLP-1 receptor agonists for glycemic control and cardiovascular health. PROSPERO REGISTRATION: CRD42024585971.

- Food-induced small bowel obstruction observed in a patient with inappropriate use of semaglutide. (Diabetology international; 2024 Oct; 10.1007/s13340-024-00751-4) — PMID 39469548 (<https://pubmed.ncbi.nlm.nih.gov/39469548/>)
- Abstract: We herein report a case of food-induced small bowel obstruction (FIBO) while using a glucagon-like peptide 1 receptor agonist (GLP1-RA), trying to lose weight due to distorted body image. The patient was a 30-year-old woman who was not obese (height 158 cm, weight 50 kg). She started taking an oral semaglutide, a GLP1-RA, and it was soon switched to weekly subcutaneous semaglutide because of ineffectiveness. More than 6 months after titrating up to 1.0 mg, she got drunk and chomped on a lot of scallops sandwiched between sheets of kelp, so-called "kobujime" in Japan, and half a day later complained of abdominal pain. Based on a finding of computed tomography at our emergency department, she was suspected of having a bowel obstruction and underwent laparoscopic surgery, which resulted in a diagnosis of small bowel obstruction by kelp. FIBO is rare, but it can become very severe once it happens. Although we cannot prove the direct pathophysiological effects of GLP1-RAs on FIBO in this particular case, GLP1-RAs

have been reported to be one of the underlying risks of bowel obstruction based on epidemiological and basic research evidence; still, it is under-recognized. For example, the package inserts in Japan do not mention intestinal obstruction. We hope that the present report will prove helpful in paying attention to GLP1-RAs as a factor in bowel obstruction, including FIBO.

- Subocclusive ileus following off-label high-dose semaglutide use. (Endocrinology, diabetes & metabolism case reports; 2025 Oct 01; 10.1530/EDM-25-0081) — PMID 41294925 (<https://pubmed.ncbi.nlm.nih.gov/41294925/>)
- Abstract: SUMMARY: Semaglutide, a GLP-1 receptor agonist (RA), is approved for the treatment of type 2 diabetes and obesity. However, its increasing off-label use, particularly for cosmetic weight loss, raises concerns considering its potential adverse effects. We report the case of a 37-year-old woman with normal body mass index who developed subocclusive ileus following unsupervised use of semaglutide.

LEARNING POINTS: Our case highlights that semaglutide can lead to functional intestinal obstruction. Clinicians should maintain a high index of suspicion for GLP-1 RA related gastrointestinal complications.

- A real-world disproportionality analysis of semaglutide: Post-marketing pharmacovigilance data. (Journal of diabetes investigation; 2024 Oct; 10.1111/jdi.14229) — PMID 38943656 (<https://pubmed.ncbi.nlm.nih.gov/38943656/>)
- Abstract: AIM/INTRODUCTION: The recent adverse reactions associated with semaglutide have led the Food and Drug Administration (FDA) to issue a "black box warning", and it is necessary to analyze all reports of adverse reactions to improve the safety of its clinical use.

MATERIALS AND METHODS: Statistical analyses and signal mining were performed by obtaining the adverse event reports related to semaglutide in the FAERS database from the first quarter of 2018 to the fourth quarter of 2023. We used disproportionality and Bayesian analysis to examine clinical and demographic attributes, trends reported quarterly, and contrasts between two distinct indications (obesity and type 2 diabetes). **RESULTS:** We found 10 unexpected adverse signals related to "pancreatic cancer", "intestinal obstruction", "cholecystitis", and "polycystic ovary" and both the two different indications had the same serious adverse reaction events occurring. **CONCLUSIONS:** This study identified many unexpected signals of serious adverse reactions, suggesting the importance of continuous post-marketing surveillance of semaglutide to understand its potential risks.

- Incretin-Based Drugs for Obesity: Common and Drug-Specific Reporting Patterns of Adverse Drug Reactions-A Comparative Disproportionality Analysis Using EudraVigilance Reports Integrating SmPC Data. (Pharmaceuticals (Basel, Switzerland); 2026 May 31; 10.3390/ph19060876) — PMID 42356493 (<https://pubmed.ncbi.nlm.nih.gov/42356493/>)
- Abstract: Background: With the increasing widespread use of GLP-1 RA and dual GIP/GLP-1 RAs in the treatment of obesity, their safety profile remains a concern for healthcare professionals (HPs). Objective: This study aimed to characterize and evaluate safety data from the EudraVigilance (EV) database for semaglutide (SEM), liraglutide (LIR), and tirzepatide (TIR). Methods: A hierarchical pharmacovigilance approach was applied, integrating SOC- and PT-level analyses with SmPC-based evaluation and both frequentist (ROR, 95% CI) and Bayesian (IC025) disproportionality methods. Within each molecule, reporter type-stratified analyses were performed, while across all molecules, disproportionality analyses were conducted separately in HP reports and in the full database to identify reporting patterns and potential safety signals, including those not described in the SmPCs. Results: Some ADRs, listed in the SmPC of only one or two of the three GLP1-RAs were also reported in the EV database for the other agents whose SmPCs do not specify these ADRs including optic ischemic neuropathy (TIR: 0.28% and LIR: 0.17%), alopecia (LIR: 0.81%), headache (TIR: 2.51%), intestinal obstruction (TIR: 1.55%), angioedema (LIR: 0.19%), hypersensitivity (SEM: 0.58% and LIR: 0.73%), etc. Pancreatitis, in particular, showed a significant but low-magnitude signal, being more frequently reported by HPs compared with non-HPs across all three GLP1-RAs. Additionally, statistically significant signals (IC025 > 0) were observed in both the HPs and full datasets. For example, for SEM vs. TIR, signals were identified for optic ischemic neuropathy (0.17; 0.13), gallbladder disorder (0.09; 0.11), and dysesthesia (0.42; 0.43), respectively. For TIR vs. SEM, signals were observed for injection site erythema (0.05; 0.11), injection site pruritus (0.01; 0.11), and injection site reaction (0.02; 0.08). Conclusions: These findings suggest potential safety signals beyond current SmPC information, emphasizing the need for continuous pharmacovigilance and cautious interpretation of reporting biases.

Limitations

- FAERS reports are spontaneous adverse event reports and do not establish causality.
- PRR and ROR describe reporting disproportionality, not incidence or patient risk.
- This openFDA analysis includes all reports mentioning the selected drug; it is not restricted to primary-suspect drugs.
- openFDA returns the latest report version, but separate consumer and sponsor submissions describing the same event may remain.
- Drug names are matched against reported and harmonized exact-name fields; incomplete normalization can omit relevant reports.
- PubMed synthesis, when requested, is restricted to the displayed retrieved abstracts and requires human review.